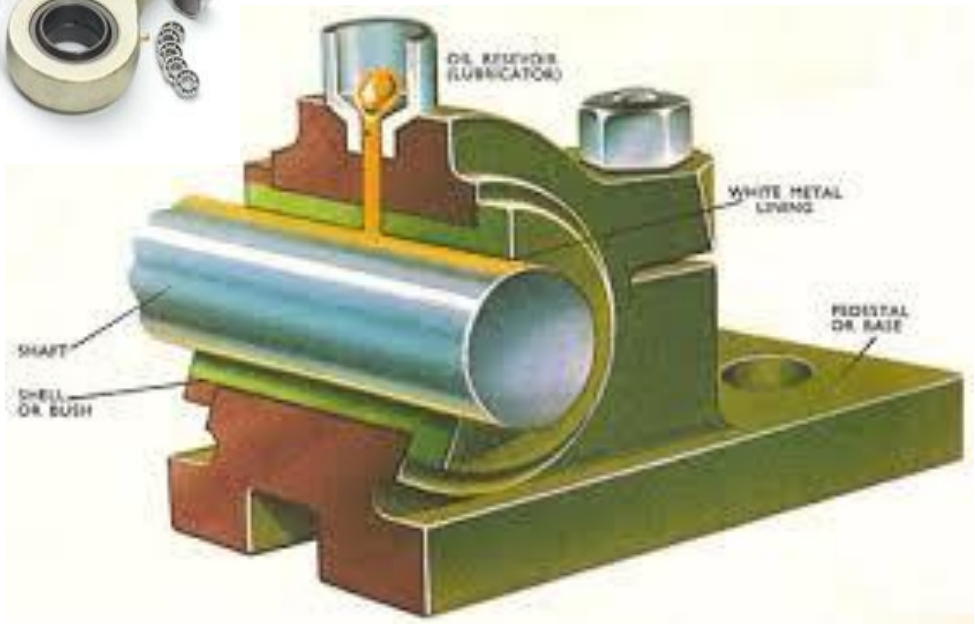
BEARINGS




OIL RESERVOIR (LUBRICATOR)
WHITE METAL LINING
SHAFT
SHELL OR BUSH
PEDESTAL OR BASE

BEARING

DEFINITION:

- A bearing is a system of machine elements whose function is to support a second member, preventing its motion in the direction of an applied load at the same time, allowing motion in another predetermined direction.

the part of a shaft or axle in contact with or enclosed by a bearing

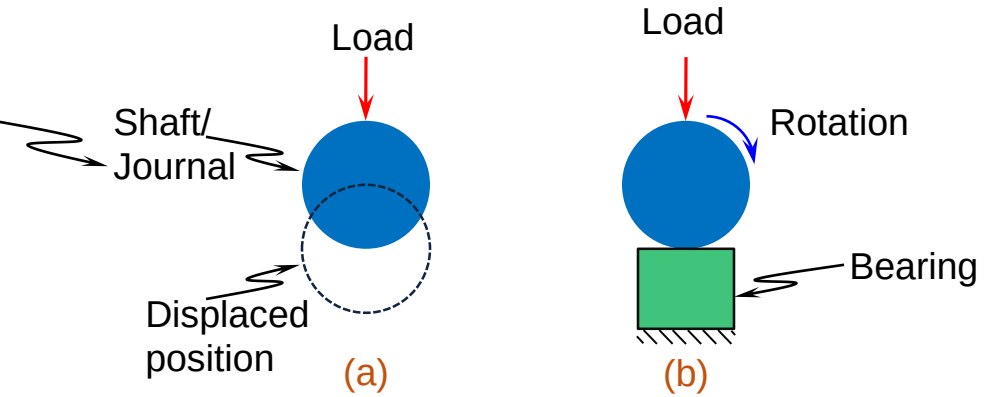


Fig. 1 End views of machine elements (a) without support and (b) with support

- *A **bearing** is a machine element that constrains relative motion to only the desired motion and reduces friction between moving parts.*
- Bearings are classified according to the direction of applied load.
  - If the bearing supports a radial load, it is called radial or shaft or sliding or journal bearing.
  - If the bearing supports a axial or thrust load, it is called thrust bearing.
  - Some bearings can support both radial and axial loads, and they are known as conical bearings.

TYPES OF BEARINGS:

- There are two main types of bearings commonly used in practice and are;
  - (a) Rolling contact bearing
  - (b) Sliding or Fluid-film bearing

- Rolling contact bearings are bearings in which balls, rollers or needless are used to support the applied load.

- Since rolling friction is lower than the sliding friction, hence rolling contact bearings have much wider use in industries.

- If the bearings are operating on the principle of hydrodynamic lubrication condition, such bearings are called fluid-film bearings.

- Fluid-film bearings are further categorized as;
  - (i) Self-acting or hydrodynamic bearings
  - (ii) Externally pressurized or hydrostatic bearings
  - (iii) Squeeze-film or elasto-hydrodynamic bearing

- A fluid-film bearing operating on the hydrodynamic condition and the load is supported due to wedging effect of the fluid caused by the relative tangential motion between two surfaces, such bearings are called self-acting bearings.

- In an externally pressurized (hydrostatic) bearing, the load is supported due to pressure in the fluid, which is supplied from an external source.

- In the case of squeeze-film bearing, which supports a load because of oscillating relative normal motion.

- In the last two categories i.e., externally-pressurized and squeeze-film bearings, the pressure within the clearance space can be generated without the wedging effect.

- Fig. 2 depicts the types of the bearings.

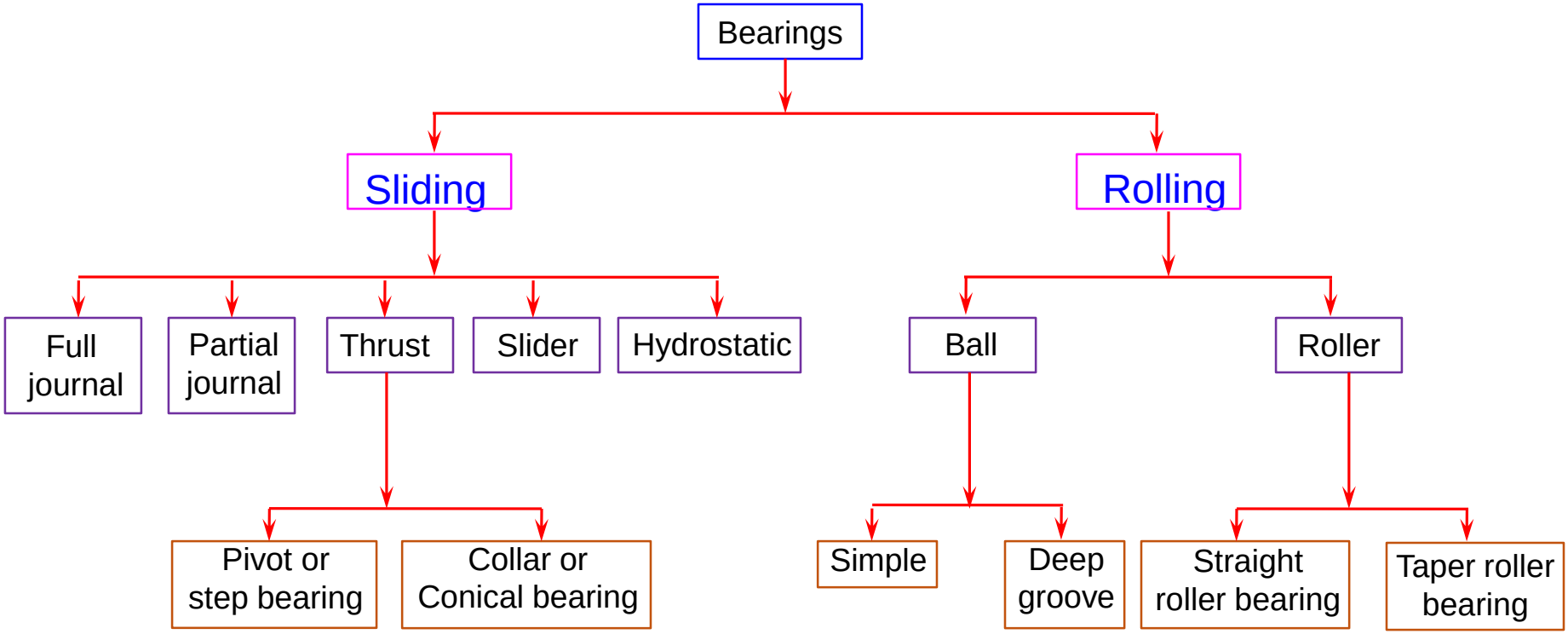


Fig. 2 Types of bearings

Journal Bearings

- Journal or shaft bearings which support radial load may be divided into full-journal bearings, where the contact angle of the bushing with the journal is 360° , and the partial-journal bearing in which the contact angle is either 180° or less.

- The shaft member of this bearing unit is called journal and the cylindrical body around the journal is called bearing.

- In most installations, the journal is a rotating element and the bearing element is stationary. Some times both elements are allowed to rotate.

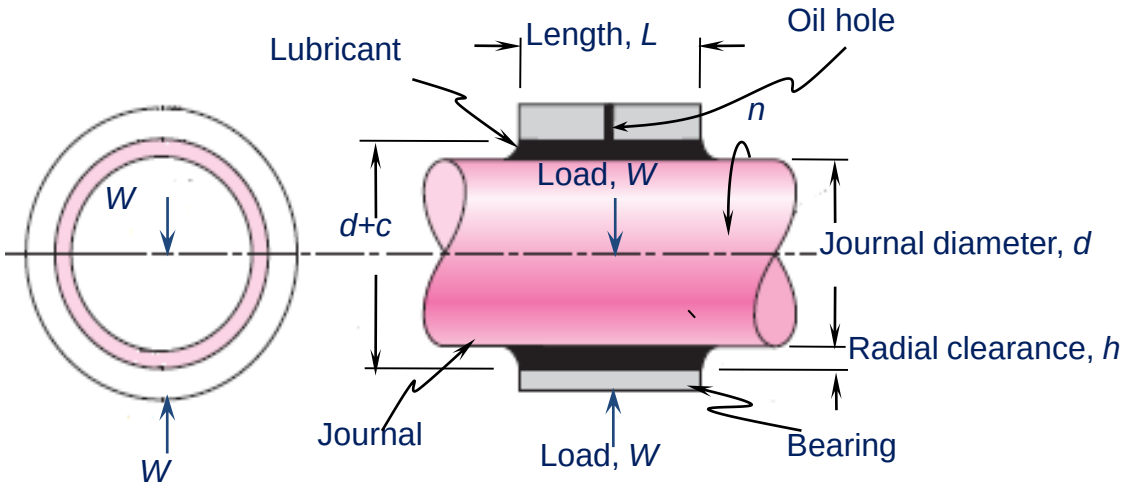
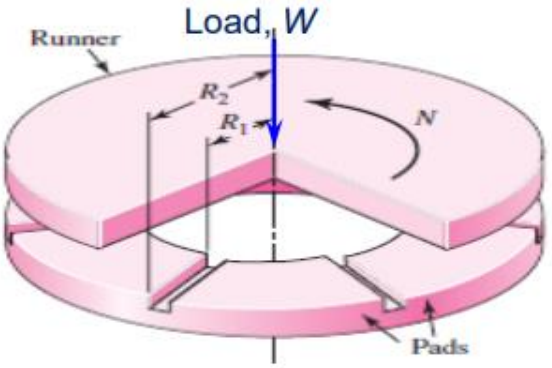


Fig. 3 Schematic of full-journal bearing

Fig. 4 represents a plane thrust bearing, in which the journal and bearing have plane rubbing surfaces, and the bearing operates under the condition of boundary lubrication.

This type of bearing is able to support only a relatively light load at moderate speeds.



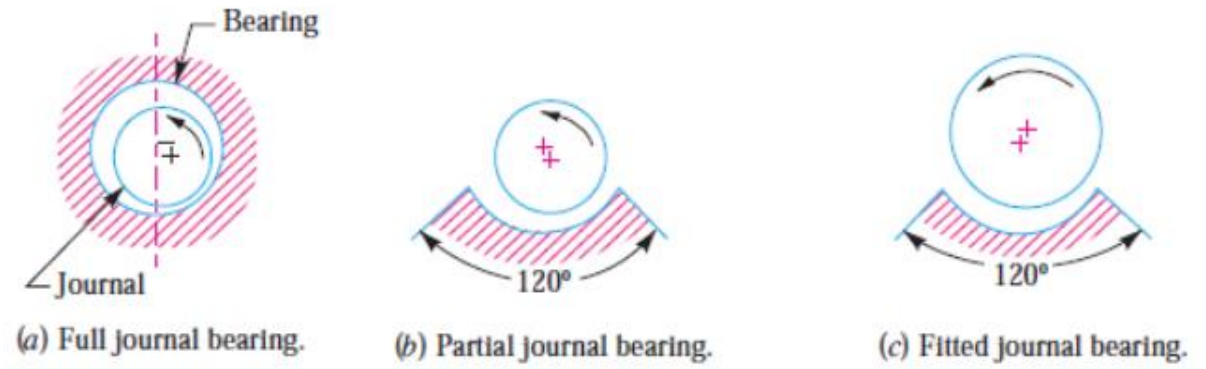


Fig. 4 Plane thrust bearing